

Chendi Li

Ph.D. student | AI Systems & HPC

homepage | chendi.li@utah.edu | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

I am currently a fourth-year PhD student at the University of Utah, advised by Prof. **P. (Saday) Sadayappan**. I'm working on the software system design for accelerating GPU kernels. I received my master's degree from the State Key Laboratory of Computer Architecture, Institute of Computing Technology, Chinese Academy of Sciences, under the supervision of Prof. **Yunquan Zhang**.

EDUCATION

Ph.D student, Computer Science , University of Utah	Sep 2022 — Now
M.S., Computer Software and Theory , University of Chinese Academy of Sciences	Sep 2019 — July 2022
B.Eng., Computer Science , Hunan Agricultural University	Sep 2014 — June 2018

PUBLICATIONS

[CVPR 2025] Cheng Yang*, Yang Sui*†, Jinqi Xiao, Lingyi Huang, Yu Gong, **Chendi Li**, Jinghua Yan, Yu Bai, Ponnuswamy Sadayappan, Xia Hu, Bo Yuan. TopV: Compatible Token Pruning with Inference Time Optimization for Fast and Low-Memory Multimodal Vision Language Model.

[ICS 2024] **Chendi Li***, Yufan Xu*, Sina Mahdipour Saravani, Saday Sadayappan, Accelerated Auto-Tuning of GPU Kernels for Tensor Computations.

[TPDS 2024] Cunyang Wei, Haipeng Jia, Yunquan Zhang, Jianyu Yao, **Chendi Li**, Wenxuan Cao. IrGEMM: An Input-Aware Tuning Framework for Irregular GEMM on ARM and X86 CPUs.

[ICS 2023] T Chen, H Jia, Y Zhang, K Li, Z Li, X Zhao, J Yao, **Chendi Li**, et al. OpenFFT: An Adaptive Tuning Framework for 3D FFT on ARM Multicore CPUs.

[ISPA 2021] **Chendi Li**, Haipeng Jia, Hang Cao, et al. AutoTSMM: An Auto-tuning Framework for Building High-Performance Tall-and-Skinny Matrix-Matrix Multiplication on CPUs.

[ICPADS 2021] Jianyu Yao, Boqian Shi, Chunyang Xiang, Haipeng Jia, **Chendi Li**, et al. IAAT: An Input-Aware Adaptive Tuning framework for Small GEMM.

[HPCC 2021] Tun Chen, Haipeng Jia, Zhihao Li, **Chendi Li**, et al. A Transpose-free Three-dimensional FFT Algorithm on ARM CPUs.

EXPERIENCES

Research Intern — Distributed GEMM Kernel Optimization	May 2025 — Aug 2025
ByteDance Seed, Bellevue, WA (Onsite)	

- Engineered a Triton-distributed kernel configuration space infrastructure for generating the distributed kernel schedules.
- Implemented a fine-grained distributed GEMM kernel for better communication and computation overlap.
- Applied the existing reinforcement learning(RL) Verl framework and completed the integration of Triton-distributed schedules.
- Built an RL-based LLM as the autotuner (Qwen3 + GRPO with Verl) that generates optimal Triton-distributed schedules, achieving consistent speed-ups over PyTorch+NCCL distributed GEMM baselines on multi-GPUs.

Graduate Student Research Assistant - GPU kernels Optimization	Sep 2022 — Now
University of Utah	

- Working on automated GPU kernels optimization, implemented analytical features and a gradient descent based autotuner, demonstrates an order-of-magnitude improvement in auto-tuning time to achieve the same level of code performance.
- Applied analytical feature + proxy-model workflow to improve kernel auto-tuning pipeline: raising random forest model accuracy from 30% (baseline) to 80% on held-out tuning samples and substantially improving sample efficiency of Bayesian search.
- Co-designed and implemented Pareto-based vector-search to accelerate convergence of GPU kernels tuning, improving overall tuning convergence by 2x to achieve comparable performance in practical benchmarks.

- Designed and developed AutoTSMM, an auto-tuning framework for irregular-shaped GEMM; outperformed Arm Compute Library, and achieved performance competitive with Intel OneMKL.
- Contributed to OpenBLAS and OpenFFT libraries, optimized OpenBLAS kernels (pre-packed GEMM, TRSM) for ARMv8 and x86.

- Implemented multi-threading and 2D-FFT optimizations in OpenFFT, improving small-scale FFT throughput and enabling broader vendor adoption.

PROJECTS

LLM for Distributed GEMM Optimization

Feb 2023 — May 2025

- Engineered an LLM-powered autotuner on a Triton-distributed framework, using Verl framework and integrated reinforcement learning to optimize schedules and achieve consistent speed-ups over PyTorch+NCCL distributed GEMM baselines on multi-GPUs.

Apache TVM, Ansor-AF-DS

Feb 2023 — May 2025

- Designed and developed the dynamic gradient algorithm in TVM. Experimental evaluation on a number of matrix-matrix multiplication and 2D convolution kernels demonstrates an order-of-magnitude improvement in auto-tuning time to achieve the same level of code performance. The paper was published in ICS'24.

TopV: Compatible Token Pruning for VLMs

2024 — 2024

- Co-authored *TopV*, a training-free, optimization-based visual token pruning method compatible with FlashAttention and KV cache; performs one-shot pruning during prefilling and maintains a reduced token set throughout decoding.

OpenBLAS,

Nov 2020 — June 2022

- Developed and optimized pre-packed matrix-matrix multiplication and triangular solve with multiple right-hand sides for ARMv8 and x86 platforms. OpenBLAS is one of the most widely used open-source BLAS libraries.

AutoTSMM,

Nov 2020 — June 2022

- AutoTSMM is a auto-tuning framework for high-performance irregular-shaped matrix multiplication on CPUs. This work was published in ISPA'21 and TPDS'24.

IAAT,

Nov 2020 — Feb 2022

- IAAT is a template-driven, just-in-time small GEMM framework targeting CPUs. This work was accepted by ICPADS'21.

AutoFFT/OpenFFT,

Jan 2018 — Feb 2021

- AutoFFT/OpenFFT is a template-based FFT code auto-generation framework. This work was published in SC'19, TPDS'20, HPCC'21, and ICS'23.

AWARDS & HONORS

- 2021
- First-class scholarships
- 2015
- Collegiate programming contest first prize

TECHNICAL SKILLS

Languages	C, C++, Python, CUDA, Triton, ARM/X86 Assembly
Parallel/Distributed	Triton-distributed, OpenMP, Pthreads, x86/ARM intrinsic
ML Compilers	PyTorch, TVM
Tools	Nsight Compute, GDB, perf, CMake, Git, Linux